

International Institute of Information Technology, Hyderabad
(Deemed to be University)

MA4.101-Real Analysis (Monsoon-2025)

Mid-Semester Exam

Time: 90 Minutes

Total Marks: 40

General Guidelines

- Attempt any four questions only.
- One A4 cheat sheet (both sides) is allowed.
- Unless stated otherwise, standard theorems from class may be used.
- Show all steps clearly; unsupported answers may not receive full credit.

Question (1) [10 Marks] Answer the following.

(a) [3 Marks] Prove that for all natural numbers $n \geq 0$,

$$(3 + \sqrt{5})^n + (3 - \sqrt{5})^n$$

is an even number.

(b) [3 Marks] Consider $c \in \mathbb{R}$. Let $S_1 \subset \mathbb{R}, S_2 \subset \mathbb{R}$ be two neighbourhoods of c . Then prove that $S_1 \cup S_2$ is a neighbourhood of c .

(c) [4 Marks] Let $S \subset \mathbb{R}$. Then S is an open set iff $S = \text{int}(S)$, where $\text{int}(S)$ is the set of all interior points of S .

Question (2) [10 Marks] Answer the following.

(a) [3 Marks] Let $N \geq 1$ be an integer. Show that there exists an integer m with

$$\frac{m}{N} \leq \sqrt{2} < \frac{m+1}{N}.$$

Deduce that $0 \leq \sqrt{2} - \frac{m}{N} < \frac{1}{N}$.

(b) [3 Marks] Conclude that $\sqrt{2}$ can be approximated arbitrarily well by rationals of the form m/N with $m \in \mathbb{Z}$, and hence that $\mathbb{Q}$ is dense in $\mathbb{R}$.

(c) [4 Marks] For $N = 50$, produce an explicit rational with denominator 50 lying below $\sqrt{2}$ and verify the error bound $< 1/50$ using inequalities.

Question (3) [10 Marks] Answer the following.

(a) [3 + 3 Marks] Consider two sequences (s_n) and (t_n) such that $\lim_{n \rightarrow \infty} s_n = s$ and $\lim_{n \rightarrow \infty} t_n = t$. Then prove that

(i) $\lim_{n \rightarrow \infty} (s_n + t_n) = s + t.$

(ii) $\lim_{n \rightarrow \infty} (s_n t_n) = st.$

(b) [4 Marks] Let $(u_n), (v_n), (w_n)$ be three sequences of real numbers and $\forall n \in \mathbb{N}$, suppose

$$u_n < v_n < w_n.$$

If $\lim_{n \rightarrow \infty} u_n = \lim_{n \rightarrow \infty} w_n = l$, then prove that $\lim_{n \rightarrow \infty} v_n = l$.

Question (4) [10 Marks] Let $(x_n)_{n \geq 1}$ be a sequence and define

$$y_n = \frac{x_n}{x_{n+1}} \quad (n \geq 1). \tag{1}$$

Answer the following.

(a) [2 Marks] Suppose the sequence (x_n) is convergent and $\lim_{n \rightarrow \infty} x_n = L$ with $L \in \mathbb{R}$. Can one conclude that $\lim_{n \rightarrow \infty} y_n = 1$? Prove it if yes, or give the correct necessary condition(s) if not.

(b) [3 Marks] Give an example of $(x_n)_{n \geq 1}$ such that $\lim_{n \rightarrow \infty} x_n = 0$ but $\lim_{n \rightarrow \infty} y_n = -1$.

(c) [3 Marks] Give another example of $(x_n)_{n \geq 1}$ such that $\lim_{n \rightarrow \infty} x_n = 0$ but $(y_n)_{n \geq 1}$ does not converge to any finite value.

(d) [2 Marks] Suppose instead that (x_n) is monotonically increasing and $x_n \rightarrow L \neq 0$. Do your conclusions change?

Question (5) [10 Marks] Consider the sequence $(a_n)_{n \geq 1}$ with $a_n = \frac{(-1)^n}{2}$. Answer the following questions.

(a) [1 Mark] Does the sequence $(a_n)_{n \geq 1}$ converge?

(b) [5 Mark] Let $L \geq 1/2$ be a positive rational number. Define

$$d_n = \inf_{k \geq n} |a_k - L|.$$

Does the sequence $(d_n)_{n \geq 1}$ converge? If no, why? If yes, what is $\lim_{n \rightarrow \infty} d_n$?
What are the limit points of the sequence $(d_n)_{n \geq 1}$?

(c) [4 Mark] Define

$$e_n = \min_{1 \leq k \leq n} |a_k - L|.$$

Does the sequence $(e_n)_{n \geq 1}$ converge? If no, why? If yes, what is $\lim_{n \rightarrow \infty} e_n$?

Question (6) [10 Marks] Define the sequence $(a_n)_{n \geq 1}$ by $a_1 = 4$ and

$$a_n = \begin{cases} -1 + \frac{1}{k}, & \text{if } n = k! \text{ for some integer } k \geq 2, \\ 1 - \frac{1}{n}, & \text{otherwise.} \end{cases}$$

Here $k! = k \times (k-1) \times \cdots \times 2 \times 1$ for $k \geq 2$. Answer the following questions.

(a) [1 Mark] Write out the first six terms $a_1, \dots, a_6$.

(b) [2 Marks] Show that the sequence $(a_n)_{n \geq 1}$ is both bounded from below and bounded from above.

(c) [3 Marks] Compute $\sup(a_n)_{n \geq 1}$ and $\inf(a_n)_{n \geq 1}$.

(d) [4 Marks] Compute $\limsup_{n \rightarrow \infty} a_n$ and $\liminf_{n \rightarrow \infty} a_n$.

Question (7) [10 Marks] A sequence $(x_n)_{n \geq 1}$ of real numbers is said to be *monotone increasing* if $x_{n+1} \geq x_n$ for all $n \geq 1$. We say $(x_n)_{n \geq 1}$ is *quasi-monotone increasing* if $x_{n+1} \geq x_n - \frac{1}{6}$ for all $n \geq 1$. Answer the following.

(a) [2 Marks] Prove that every monotone increasing sequence is quasi-monotone increasing as well.

(b) [4 Marks] Let $x_n = 1 + \frac{(-1)^n}{18n}$ and

$$y_n = \begin{cases} 0 & \text{if } n \text{ is even} \\ \frac{1}{18} \left(1 + \frac{1}{n}\right) & \text{if } n \text{ is odd} \end{cases}$$

Prove that $(x_n)_{n \geq 1}$ and $(y_n)_{n \geq 1}$ are quasi-monotone increasing sequences that are not monotone increasing.

(c) [2 Marks] Prove or disprove: every bounded quasi-monotone increasing sequence converges.

(d) [2 Marks] Compare this with the Monotone Convergence Theorem. What essential feature of true monotonicity is lost in the quasi-monotone case, and why does this allow bounded quasi-monotone sequences to fail to converge?

Question (8) [10 Marks] Prove that in $\mathbb{R}$

(a) [5 Marks] Every Cauchy sequence is convergent.

(b) [5 Marks] Every convergent sequence is Cauchy.